US Electric Vehicle Adoption & Renewable Energy Analysis Dashboard

Project Report

Prepared By

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Tools Used

- Power BI
- Power Query
- DAX
- Excel
- Data Visualization

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1. Abstract

The increasing adoption of Electric Vehicles (EVs) and the expansion of renewable energy sources are transforming the transportation and energy sectors. This project analyzes the relationship between EV adoption and renewable energy production across the United States. Using Power BI, multiple datasets related to vehicle registrations and energy generation were integrated, cleaned, and visualized to identify patterns and trends.

The objective of this project is to determine whether states with higher renewable energy production also exhibit higher levels of electric vehicle adoption. The findings provide valuable insights into the connection between sustainable energy infrastructure and transportation electrification.

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2. Introduction

Electric Vehicles have emerged as an important solution for reducing greenhouse gas emissions and dependence on fossil fuels. At the same time, renewable energy sources such as solar, wind, and hydroelectric power are becoming increasingly significant in electricity generation.

This project investigates state-level EV adoption and renewable energy production data to understand the relationship between clean energy availability and EV growth.

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3. Problem Statement

The main objective of this project is:

"To analyze whether states with higher renewable energy production demonstrate higher electric vehicle adoption rates."

Key Questions:

- Which states have the highest EV registrations?
- Which states generate the highest percentage of renewable energy?
- Is there a relationship between renewable energy share and EV adoption?
- What recommendations can be made to encourage sustainable transportation?

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4. Objectives

The objectives of this project are:

1. Analyze EV adoption across US states.
2. Evaluate renewable energy production trends.
3. Compare renewable and fossil energy generation.
4. Identify relationships between renewable energy and EV adoption.
5. Generate actionable insights and recommendations.

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5. Dataset Description

The project combines multiple datasets:

EV Registration Dataset

Contains state-wise electric vehicle registrations for 2018.

Vehicle Registration Dataset

Contains total registered vehicles by state.

Energy Generation Dataset

Contains electricity generation data from multiple energy sources including:

- Coal
- Natural Gas
- Petroleum
- Nuclear
- Hydro
- Wind
- Solar
- Geothermal
- Biomass

State Codes Dataset

Used for standardization and geographic visualization.

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6. Data Preparation

The following data preparation steps were performed:

Data Cleaning

- Removed duplicate records.
- Removed blank values.
- Standardized state names.
- Corrected data types.

Data Transformation

- Merged datasets using state identifiers.
- Created EV adoption percentage metrics.
- Calculated renewable energy percentages.
- Categorized states into renewable energy groups.

Data Modeling

Relationships were established to support dashboard visualizations and KPI calculations.

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7. Methodology

The project followed the following workflow:

Data Collection

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Data Cleaning

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Data Transformation

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Data Modeling

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Dashboard Development

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Analysis & Insights

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Recommendations

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8. Dashboard Overview

The Power BI dashboard consists of five pages:

Page 1: Executive Summary

Provides a high-level overview of:

- Total EV Registrations
- Total Vehicles
- Number of States
- Average Renewable Energy Share

Page 2: EV Adoption Analysis

Focuses on:

- Top EV states
- EV adoption rates
- Geographic distribution of EVs

Page 3: Renewable Energy Analysis

Focuses on:

- Renewable vs Fossil energy production
- Top renewable states
- Energy source contributions

Page 4: Correlation Analysis

Analyzes the relationship between:

- Renewable Energy Share
- EV Adoption Rate

Page 5: Key Insights & Recommendations

Summarizes major findings and recommendations.

9. Analysis and Findings

EV Adoption Analysis

The analysis revealed that a small number of states account for a significant share of EV registrations.

Key observations:

- California leads EV registrations.
- EV adoption is concentrated in a limited number of states.
- States with strong environmental policies generally show higher EV adoption.

Renewable Energy Analysis

Renewable energy production varies significantly across states.

Key observations:

- Wind and Solar contribute substantially to renewable generation.
- Several states rely heavily on renewable sources.
- Fossil fuel dependence remains significant in some regions.

Correlation Analysis

A scatter plot was used to evaluate the relationship between renewable energy production and EV adoption.

Key observations:

- States with higher renewable energy percentages generally show stronger EV adoption.
- Clean energy infrastructure appears to support transportation electrification.
- Renewable energy and EV adoption exhibit a positive relationship in many states.

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10. Key Findings

1. California recorded the highest EV registrations among all states.

2. States with higher renewable energy shares generally exhibited higher EV adoption rates.

3. Wind and Solar energy were major contributors to renewable electricity generation.

4. States heavily dependent on fossil fuels showed comparatively lower EV adoption.

5. Renewable energy infrastructure appears to support EV ecosystem growth.

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11. Recommendations

1. Increase investment in renewable energy projects.
2. Expand EV charging infrastructure.
3. Continue EV incentives and subsidies.
4. Promote sustainable transportation policies.
5. Encourage integration of renewable energy with EV charging networks.

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12. Conclusion

This project demonstrates that renewable energy production and electric vehicle adoption are closely connected in many US states. States with stronger renewable energy infrastructure generally show higher levels of EV adoption.

The findings highlight the importance of investing in renewable energy and EV infrastructure to support long-term sustainability goals. Policymakers and stakeholders can use these insights to accelerate the transition toward clean transportation and low-carbon energy systems.

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13. Future Scope

Future enhancements may include:

- Multi-year trend analysis.
- EV growth forecasting.
- Renewable energy forecasting.
- Advanced statistical correlation analysis.
- Machine learning models for EV adoption prediction.

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14. References

- U.S. Energy Information Administration (EIA)
- State Vehicle Registration Statistics
- Public Renewable Energy Datasets
- Power BI Documentation

Final Statement

Renewable Energy and Electric Vehicles together play a critical role in building a sustainable, low-carbon, and environmentally responsible future.